DO NOW — RECALL

Hash tables with arrays

Name: _____ Class: _____ Date: _____

1. In a binary search tree, where are values smaller and larger than a node stored?

2. What does -1 represent in our pointer arrays?

3. What is a collision in hashing?

4. For table size 5, what bucket does key 17 map to using key MOD 5?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Smaller values go left and larger values go right.

2. Model answer

null / no child.

3. Model answer

Two keys map to the same hash location.

4. Model answer

2.

By the end, you can…

• explain how a hash function maps keys to buckets

• handle collisions using fixed array buckets

• create, traverse, add, search for and remove hashed data

Vocabulary — know these terms

Recall the term — hash table

Recall the term — key

Recall the term — hash function

Recall the term — bucket

Recall the term — collision

Recall the term — chaining

Recall the term — lookup

Recall the term — load

NEW LEARNING

Read, annotate and explain

Hash chooses the row; collision uses another slot

table size = 5
hash(key) = key MOD 5

12 → row 2
17 → row 2
22 → row 2

bucket 2: [12, 17, 22]

A fixed 2D array lets us model separate chaining without introducing OO or library collection classes.

Create fixed buckets

Each row is a bucket with three available collision slots.

Hash

The same key maps to the same row.

Add

A collision goes into the next free column in that bucket.

Search/remove

Only the calculated bucket is searched. Removal can shift later bucket items left.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Add 12, 17 and 22

12

```
table[2,0] = 12
count[2] = 1
```

17

```
table[2,1] = 17
count[2] = 2
```

22

```
table[2,2] = 22
count[2] = 3
```

17

```
hash → bucket 2
scan row 2
17 found at column 1
```

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Calculate hash

Table size 10; key 37.

```
37 MOD 10 = ?
```

Method / working:

2. Add key

Use bucket and count arrays.

```
int bucket = key % ?;
table[bucket, ?] = key;
count[bucket]++;
```

Method / working:

3. Collision

12 and 17 both map to bucket 2.

```
bucket 2:
column 0 = 12
column 1 = ?
```

Method / working:

4. Remove from bucket

Remove column 1 when count is 3.

```
table[b,1] = table[b,?];
count[b]--;
```

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Create buckets
Create a 5×3 integer table and a count array.

Question 2

Add keys with collisions
Add 12, 17, 22 and 9 using key MOD 5.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Search only the correct bucket
Search for key 17 using its hash value and a loop across that bucket.

Question 4

Remove 17
Find 17 in its bucket, shift later values left, and reduce the bucket count.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A hash function tells you which bucket to search; it does not guarantee that only one key will use that bucket.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.